



ENGINEERING GUIDE

Engineering Toolkit

Repeatable analysis, camera, tracing, and native-research workflows.

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The VirtualGlove Engineering Toolkit is an optional, version-matched collection for people who want to measure, reproduce, or extend the camera-to-game pipeline. It is not required to install VirtualGlove, personalize a hand, adjust movement reach, or play a game.

The toolkit favours repeatable evidence over live guesswork. Most investigations begin with an existing trace or recorded clip, proceed through an output-free comparison, and reach a running Controller only after the offline result is understood.

Pick a workflow

You do not need to learn every tool. Start with the question you are trying to answer and use the smallest workflow that can answer it.

Question	Start here	Continue only when needed
Did recognition change?	guided-vision-benchmark.py , then benchmark-vision-replay.py	benchmark-palm-anchors.py or one of the isolated tracker experiments
Is movement responsive and stable?	benchmark-native-motion-curve.py	analyze-motion-trace.py or benchmark-direction-response.py
Where is time being spent?	measure-vision-status.py	prepare-end-to-end-session.py , then the bounded latency-tracing workflow
Is camera delivery the bottleneck?	benchmark-camera-pipeline.py	frame-preprocessing or exposure soaks
Does native Super Glove Ball still behave correctly?	build-nestopia-powerglove.sh , then run-nestopia-powerglove-trace.py	the matched direction-response comparison
Could another recognition backend help?	Use the retained Tasks, staggered-tracker, or ncnn comparison	Treat the result as research evidence, not a production switch

Prefer offline tools first. Commands that open a camera or temporarily restart services say so in their help and require deliberate operator steps.

Choose the right download

Download [VirtualGlove-Engineering-Tools.zip](#) from the same GitHub release as the Controller and console installations. Do not mix toolkit and device versions when comparing behaviour.

Also download [VirtualGlove-Engineering-Tools.zip.sha256](#). Verify the download before extracting it:

```
SH
shasum -a 256 -c VirtualGlove-Engineering-Tools.zip.sha256
unzip VirtualGlove-Engineering-Tools.zip
cd VirtualGlove-Engineering-Tools
```

On Linux, `sha256sum -c` can be used instead of `shasum -a 256 -c`.

First self-check

The initial check needs only Python and does not open a camera, contact a device, install packages, or send controller input:

```
SH
python3 scripts/check-engineering-toolkit.py
```

It verifies the package manifest, paths, Python syntax, direct helper files, and every supported command's side-effect-free help interface. Resolve a failed self-check before collecting evidence.

Create the local environment

Python 3.10 or newer can run offline analysis and video review. The setup script creates `.venv-engineering` inside the extracted toolkit and records the exact resolved package versions in `.venv-engineering/virtualglove-engineering-environment.json`:

```
SH
python3 scripts/setup-engineering-tools.py
```

Activate it on macOS or Linux:

```
SH
. .venv-engineering/bin/activate
```

On Windows PowerShell:

```
POWERSHELL
.\.venv-engineering\Scripts\Activate.ps1
```

MediaPipe comparisons use Python 3.12, matching the supported Controller toolchain:

```
SH
python3.12 scripts/setup-engineering-tools.py --with-mediapipe
```

Rerunning either setup is safe. Verify an existing environment without changing it by adding `--check` and the same MediaPipe option originally used. The check compares the toolkit version, setup mode, Python version, exact resolved package record, and validated direct dependency versions.

The public toolkit does not carry the Controller's ARM64 MediaPipe wheel. A Controller installation already has its validated runtime; do not replace it with a generic workstation package. Use the packaged tools from a workstation for offline analysis and orchestration, or the existing deployed engineering copy on a development Controller when direct camera access is required.

Tool map

The generated [engineering-tools.json](#) records the exact toolkit version, complete file inventory, and these categories.

Category	Main commands	Changes a running system?
Offline analysis	Latency, motion-trace, direction-response, preprocessing, movement-curve, palm-anchor, post-inference, and diagnostic-overhead tools	No
Camera and recognition	Guided capture, vision replay, camera-pipeline measurement, isolated tracker comparisons, and exposure soaks	May take exclusive ownership of the camera; never sends controller output
Live system tracing	Status measurement, end-to-end preflight, bounded trace management, and native latency sessions	Observation is read-only; trace sessions temporarily restart measured services and restore production when stopped normally
Native and accelerator research	Pinned Nestopia and FCEUmm builders, native-core trace instrumentation, ncnv, and MediaPipe Tasks comparisons	Builds only in a local destination and never installs a core; probes may run caller-supplied binaries or models
Toolkit support	Environment setup and the dependency-free package self-check	Creates or verifies files only inside the toolkit environment

Shared [src/virtualglove](#) and [config](#) source is included where tools import or inspect it. Native-emulation research receives the editable VirtualGlove patch and diagnostic header, not the platform release binaries. The package also contains the project licence, third-party notices, and this guide.

The archive intentionally excludes ROMs, recorded video, trace results, credentials, pairing tokens, player settings, personal calibration, cached models, compiled emulator cores, platform core source archives, and private device data.

Repository-only maintenance tools are excluded: release publishing, device deployment, firmware stamping, package construction, PDF and screenshot generation, and precompiled Matrix release building. The ordinary Controller package already carries the verified firmware; rebuilding it requires the pinned Arduino CLI, Zephyr platform, and libraries from [sketch/sketch.yaml](#) in the full Git repository.

Start with offline evidence

Use `--help` on every command before its first run. These examples demonstrate the safest common entry points.

Summarize a Controller motion trace:

```
SH
python scripts/analyze-motion-trace.py TRACE.json --output motion-summary.json
```

Join Controller and receiver latency traces. Add `--core` only when a matching native-core trace exists:

```
SH
python scripts/analyze-latency-trace.py \
  --controller controller.json \
  --receiver receiver.json \
  --output latency-summary.json
```

Prepare a local slow-motion video for reviewed frame annotation:

```
SH
python scripts/analyze-latency-video.py \
  --video recording.mov \
  --protocol smoke \
  --capture-fps 120 \
  --output-dir video-review
```

Replay a retained camera clip through the proven recognition lane:

```
SH
python scripts/benchmark-vision-replay.py \
  clip.avi --quick --output replay.json
```

Recorded clips and generated reports remain local unless the operator moves or uploads them. Review them for faces, rooms, hostnames, or other private material before sharing.

Read-only live observation

The status sampler observes the public Controller status endpoint without opening the camera or changing controller delivery:

```
SH
python scripts/measure-vision-status.py \
  --status-url http://CONTROLLER.local:8088/status?statistics=1 \
  --seconds 60 \
  --phase movement \
  --output status-sample.json
```

The end-to-end preflight checks a development checkout and both devices. It uses SSH but does not change them:

```
SH
python scripts/prepare-end-to-end-session.py \
  --controller-status http://CONTROLLER.local:8088/status?statistics=1 \
  --controller-ssh arduino@CONTROLLER.local \
  --retroPie-ssh pi@RETROPIE.local \
  --output-dir preflight
```

Use dedicated SSH identities through the corresponding identity options. Do not place private keys inside the toolkit directory or an evidence bundle.

Guided camera capture

Stop normal vision before allowing an engineering tool to own the camera. Keep controller output stopped and ensure no game depends on the camera during the test.

The single supported recorder provides a local live preview and user-paced cues. It replaced the older fixed-duration recorder so the operator can verify each pose instead of racing a timer:

```
SH
python scripts/guided-vision-benchmark.py \
  --camera auto \
  --protocol tracking \
  --output guided-tracking.avi
```

The page address is printed when the recorder starts. Finish or interrupt the recorder normally so it releases the camera. Restart normal vision and confirm the Dashboard preview before playing.

Camera pipeline and exposure tools require explicit acknowledgement flags. Those flags confirm that the operator has stopped the ordinary camera worker; they are not shortcuts around that safety step.

Tracing a running two-device system

[manage-latency-traces.py](#) temporarily recreates measured services with bounded diagnostic tracing. Use its [start](#) and [stop](#) subcommands as one matched pair. The stop action restores production before collecting finalized traces.

Before starting:

1. Confirm Controller and target Linux console SSH access.
2. Confirm the expected game and emulator are selected.
3. Stop if either device is already unhealthy.
4. Choose a short bounded duration.
5. Keep a separate terminal ready to run the stop command.

Do not publish raw trace bundles. Although controller packets contain no camera frames, reports can contain host information, paths, timings, and configuration details.

Native emulator research

Build the pinned, patched Nestopia core in a local directory without installing it:

```
SH
sh scripts/build-nestopia-powerglove.sh build/nestopia-research
```

Build a stock FCEUmm comparison core when a matched joystick test needs it:

```
SH
sh scripts/build-fceumm-benchmark.sh build/fceumm-research
```

Both builders fetch pinned upstream source. Review their output and licences; do not substitute their local products for platform-packaged release cores.

[run-nestopia-powerglove-trace.py](#) loads a caller-supplied libretro core and ROM without installing either one. The toolkit never supplies commercial ROMs. Keep ROM and scratch paths outside the extracted toolkit, and remove scratch copies when the run finishes.

The ncnn and MediaPipe Tasks tools are retained for reproducible comparison; they are not production alternatives. A faster isolated layer or delegate is not sufficient evidence. Any proposed runtime must preserve recognition, continuity, ordering, thermal stability, and complete camera-to-game latency.

The toolkit no longer includes the old fixed recorder, [min/boost](#) filename matrix, or earlier ideal-step sampler. Their assumptions belonged to retired motion experiments. Guided capture, labelled replay evidence, and the bounded native-motion curve are the maintained replacements.

Evidence checklist

Every retained result should record:

- VirtualGlove release and toolkit version
- Controller and console build identifiers
- Camera model, reader, frame rate, buffer count, exposure, and gain
- Python and resolved package environment record
- Exact command and input-file digests
- Whether Dashboard preview and statistics were open
- Test duration, operator cues, limitations, and unexpected interruptions
- Aggregate output filename and SHA-256 digest

Never treat results from different clips or physical movements as a controlled A/B comparison. Never combine latency percentiles measured on clocks that were not synchronized.

Cleanup and recovery

Generated environments, clips, traces, and reports are not managed by the VirtualGlove installer.

- Delete and recreate `.venv-engineering` when changing Python families.
- Keep one known-good environment record with important evidence.
- Move valuable clips and aggregate reports into a clearly named private evidence directory.
- Remove abandoned scratch directories and duplicate extracted toolkits.
- Do not clear the Controller's active `uv` cache merely to make space; it supports reliable offline worker restarts.
- Archive old development-device evidence before removing it.
- After any camera-owning experiment, confirm the Dashboard preview and normal game control before declaring recovery complete.

For active file locations and option definitions, see the online [Configuration Reference](#). For the current runtime and device boundaries, see [Architecture and flows](#). For why specific experiments were retained or rejected, see the [Engineering Journey](#).